

Enterprise Web Application Assignment 2

Assignment Tasks:

Q1. Develop a Java application for an online food delivery company that manages different types of food orders. Create packages: food.model, food.service, food.utility, food.main. Create an abstract class: FoodOrder having: orderId, customerName, amount and static member: restaurantName. Include: abstract double calculateDeliveryCharge();

Create an interface: Discountable having double applyDiscount();

Create two subclasses: RegularOrder, PremiumOrder.

The delivery charges for Regular order is Rs. 80 and for Premium is Rs. 50. The discount for Regular orders is 10% and for Premium orders is 15%.

Use: private data members, getters/setters, static object counter, static method to display total orders, constructors, method overriding concepts.

Create: OrderUtility to validate amount, validate customer name and generate order summary.

In the driver(Main) program:

- i. create at least six orders,
- ii. store them using an array of FoodOrder
- iii. display bill
- iv. display discount
- v. display delivery charge
- vi. display final payable amount

1. FoodOrder.java

```
package food.model;

public abstract class FoodOrder {
    private int orderId;
    private String customerName;
    private double amount;
    public static String restaurantName = "Grill Master";
    private static int totalOrders = 0;

    public FoodOrder(int orderId, String customerName, double amount) {
        this.orderId = orderId;
        this.customerName = customerName;
        this.amount = amount;
        totalOrders++;
    }

    public int getOrderId() {
        return orderId;
    }

    public void setOrderId(int orderId) {
        this.orderId = orderId;
    }

    public String getCustomerName() {
        return customerName;
    }

    public void setCustomerName(String customerName) {
        this.customerName = customerName;
    }

    public double getAmount() {
        return amount;
    }
```

```
}

public void setAmount(double amount) {
    this.amount = amount;
}

public static String getRestaurantName() {
    return restaurantName;
}

public static void setRestaurantName(String name) {
    restaurantName = name;
}

public static int getTotalOrders() {
    return totalOrders;
}

public static void displayTotalOrders() {
    System.out.println("\nTotal Orders = " + totalOrders);
}

public abstract double calculateDeliveryCharge();
}
```

2. Discountable.java

```
package food.service;

public interface Discountable {
    double applyDiscount();
}
```

3. RegularOrder.java

```
package food.model;

import food.service.Discountable;

public class RegularOrder extends FoodOrder implements Discountable {

    public RegularOrder(int orderId, String customerName, double amount) {
        super(orderId, customerName, amount);
    }

    @Override
    public double calculateDeliveryCharge() {
        return 80.0;
    }

    @Override
    public double applyDiscount() {
        return getAmount() * 0.10;
    }
}
```

4. PremiumOrder.java

```
package food.model;

import food.service.Discountable;

public class PremiumOrder extends FoodOrder implements Discountable {

    public PremiumOrder(int orderId, String customerName, double amount) {
        super(orderId, customerName, amount);
    }

    @Override
    public double calculateDeliveryCharge() {
        return 50.0;
    }

    @Override
    public double applyDiscount() {
        return getAmount() * 0.15;
    }
}
```

```
}
```

5. OrderUtility.java

```
package food.utility;

import food.model.FoodOrder;
import food.service.Discountable;

public class OrderUtility {

    public static boolean validateAmount(double amount) {
        return amount > 0;
    }

    public static boolean validateCustomerName(String name) {
        return name != null && !name.trim().isEmpty();
    }

    public static void generateSummary(FoodOrder order) {
        double discount = 0.0;
        if (order instanceof Discountable) {
            discount = ((Discountable) order).applyDiscount();
        }
        double delivery = order.calculateDeliveryCharge();
        double finalAmount = order.getAmount() - discount + delivery;

        System.out.println("-----");
        System.out.println("Restaurant : " + FoodOrder.restaurantName);
        System.out.println("Order ID : " + order.getId());
        System.out.println("Customer : " + order.getCustomerName());
        System.out.println("Bill : Rs. " + order.getAmount());
        System.out.println("Discount : Rs. " + discount);
        System.out.println("Delivery : Rs. " + delivery);
        System.out.println("Final Bill : Rs. " + finalAmount);
    }
}
```

6. Main.java

```
package food.main;

import food.model.FoodOrder;
import food.model.RegularOrder;
import food.model.PremiumOrder;
import food.utility.OrderUtility;

public class Main {
    public static void main(String[] args) {
        FoodOrder[] orders = {
            new RegularOrder(101, "Aarav Sharma", 500.0),
            new PremiumOrder(102, "Bhavna Patel", 800.0),
            new RegularOrder(103, "Chirag Verma", 400.0),
            new PremiumOrder(104, "Divya Kapoor", 1000.0),
            new RegularOrder(105, "Eshan Gupta", 600.0),
            new PremiumOrder(106, "Farhan Khan", 900.0)
        };

        for (FoodOrder order : orders) {
            if (OrderUtility.validateAmount(order.getAmount()) &&
                OrderUtility.validateCustomerName(order.getCustomerName())) {
                OrderUtility.generateSummary(order);
            } else {
                System.out.println("Invalid Order: " + order.getId());
            }
        }

        FoodOrder.displayTotalOrders();
    }
}
```

Q1 Output

```
shreyaas@Shreyaas-MacBook-Air Enterprise Web Applications % /opt/homebrew/opt/openjdk/bin/java -cp bin food.main.Main
-----
Restaurant : Grill Master
Order ID   : 101
Customer   : Aarav Sharma
Bill       : Rs. 500.0
Discount   : Rs. 50.0
Delivery   : Rs. 80.0
Final Bill : Rs. 530.0
-----
Restaurant : Grill Master
Order ID   : 102
Customer   : Bhavna Patel
Bill       : Rs. 800.0
Discount   : Rs. 120.0
Delivery   : Rs. 50.0
Final Bill : Rs. 730.0
-----
Restaurant : Grill Master
Order ID   : 103
Customer   : Chirag Verma
Bill       : Rs. 400.0
Discount   : Rs. 40.0
Delivery   : Rs. 80.0
Final Bill : Rs. 440.0
-----
Restaurant : Grill Master
Order ID   : 104
Customer   : Divya Kapoor
Bill       : Rs. 1000.0
Discount   : Rs. 150.0
Delivery   : Rs. 50.0
Final Bill : Rs. 900.0
-----
Restaurant : Grill Master
Order ID   : 105
Customer   : Eshan Gupta
Bill       : Rs. 600.0
Discount   : Rs. 60.0
Delivery   : Rs. 80.0
Final Bill : Rs. 620.0
-----
Restaurant : Grill Master
Order ID   : 106
Customer   : Farhan Khan
Bill       : Rs. 900.0
Discount   : Rs. 135.0
Delivery   : Rs. 50.0
Final Bill : Rs. 815.0
-----
Total Orders = 6
```

Q2. A university wants to develop a Smart Library Resource Management System to manage Books, Journals, and Digital Resources.

Create the following package structure:

- library.model
- library.service
- library.util
- library.main

Create an abstract class LibraryResource with: resourceId, title, author, static libraryName, constructor, abstract method: double calculateFine(int overdueDays);

Create an interface: Printable having method void printDetails();

Create two classes: Book, DigitalResource that extend LibraryResource and implement Printable

The fine calculation rules are: Book Rs. 5/day and Digital Resource Rs. 2/day

Use: private variables, protected methods (where appropriate), public methods, static variable, static method to display total resources created.

Maintain a static object counter that counts every resource created.

Create a utility class: InputValidator inside the utility package that validates Resource ID (>0) and Fine days (≥0)

In the main package:

- i. create at least five objects,
- ii. store them in an array,
- iii. display complete details,
- iv. calculate total fine of all overdue resources.

1. LibraryResource.java

```
package library.model;

public abstract class LibraryResource {
    private int resourceId;
    private String title;
    private String author;

    public static String libraryName = "Nava Nalanda Library";
    private static int totalResources = 0;

    public LibraryResource(int resourceId, String title, String author) {
        this.resourceId = resourceId;
        this.title = title;
        this.author = author;
        totalResources++;
    }

    public int getResourceId() {
        return resourceId;
    }

    public void setResourceId(int resourceId) {
        this.resourceId = resourceId;
    }

    public String getTitle() {
        return title;
    }

    public void setTitle(String title) {
        this.title = title;
    }

    public String getAuthor() {
        return author;
    }

    public void setAuthor(String author) {
```

```
        this.author = author;
    }

    public static String getLibraryName() {
        return libraryName;
    }

    public static void setLibraryName(String name) {
        libraryName = name;
    }

    public static int getTotalResources() {
        return totalResources;
    }

    public static void displayTotalResources() {
        System.out.println("\nTotal Resources = " + totalResources);
    }

    protected void showBasicDetails() {
        System.out.println("Resource ID : " + resourceId);
        System.out.println("Title      : " + title);
        System.out.println("Author   : " + author);
    }

    public abstract double calculateFine(int overdueDays);
}
```

2. Printable.java

```
package library.service;

public interface Printable {
    void printDetails();
}
```

3. Book.java

```
package library.model;

import library.service.Printable;

public class Book extends LibraryResource implements Printable {

    public Book(int resourceId, String title, String author) {
        super(resourceId, title, author);
    }

    @Override
    public double calculateFine(int overdueDays) {
        return overdueDays * 5.0;
    }

    @Override
    public void printDetails() {
        System.out.println("\nBook");
        showBasicDetails();
    }
}
```

4. DigitalResource.java

```
package library.model;

import library.service.Printable;

public class DigitalResource extends LibraryResource implements Printable {

    public DigitalResource(int resourceId, String title, String author) {
        super(resourceId, title, author);
    }

    @Override
    public double calculateFine(int overdueDays) {
```

```

        return overdueDays * 2.0;
    }

    @Override
    public void printDetails() {
        System.out.println("\nDigital Resource");
        showBasicDetails();
    }
}

```

5. InputValidator.java

```

package library.util;

public class InputValidator {

    public static boolean validateResourceId(int id) {
        return id > 0;
    }

    public static boolean validateFineDays(int days) {
        return days >= 0;
    }
}

```

6. Main.java

```

package library.main;

import library.model.LibraryResource;
import library.model.Book;
import library.model.DigitalResource;
import library.service.Printable;
import library.util.InputValidator;

public class Main {
    public static void main(String[] args) {
        LibraryResource[] resources = {
            new Book(1, "Operating System Concepts", "Silberschatz"),
            new DigitalResource(2, "Data Structures and Algorithms PDF", "Cormen"),
            new Book(3, "Computer Networks", "Andrew Tanenbaum"),
            new DigitalResource(4, "Java: The Complete Reference E-Book", "Herbert Schildt"),
            new Book(5, "Database System Concepts", "Korth")
        };

        int[] overdueDays = {3, 5, 2, 0, 4};
        double totalFine = 0;

        for (int i = 0; i < resources.length; i++) {
            if (InputValidator.validateResourceId(resources[i].getResourceId())
                && InputValidator.validateFineDays(overdueDays[i])) {

                ((Printable) resources[i]).printDetails();

                double fine = resources[i].calculateFine(overdueDays[i]);
                System.out.println("Overdue Days: " + overdueDays[i]);
                System.out.println("Fine      : Rs. " + fine);

                totalFine += fine;
            } else {
                System.out.println("Invalid resource data at index " + i);
            }
        }

        System.out.println("\n-----");
        System.out.println("Library Name : " + LibraryResource.libraryName);
        System.out.println("Total Fine   : Rs. " + totalFine);
        LibraryResource.displayTotalResources();
    }
}

```

Q2 Output

```
shreyaas@Shreyaass-MacBook-Air Enterprise Web Applications % /opt/homebrew/opt/openjdk/bin/java -cp bin library.main.Main

Book
Resource ID : 1
Title       : Operating System Concepts
Author      : Silberschatz
Overdue Days: 3
Fine        : Rs. 15.0

Digital Resource
Resource ID : 2
Title       : Data Structures and Algorithms PDF
Author      : Cormen
Overdue Days: 5
Fine        : Rs. 10.0

Book
Resource ID : 3
Title       : Computer Networks
Author      : Andrew Tanenbaum
Overdue Days: 2
Fine        : Rs. 10.0

Digital Resource
Resource ID : 4
Title       : Java: The Complete Reference E-Book
Author      : Herbert Schildt
Overdue Days: 0
Fine        : Rs. 0.0

Book
Resource ID : 5
Title       : Database System Concepts
Author      : Korth
Overdue Days: 4
Fine        : Rs. 20.0

-----
Library Name : Nava Nalanda Library
Total Fine   : Rs. 55.0

Total Resources = 5
```